

sleep操作

注意

修改的每个文件都要保存好

编写sleep的源代码

```
//路径 /user/sleep.c
#include "kernel/types.h"
#include "kernel/stat.h"
#include "user/user.h"

int
main(int argc, char *argv[])
{
    if(argc < 2){
        fprintf(2, "Usage: sleep number\n");
        exit(1);
    }
    int number = atoi(argv[1]);
    sleep(number);
    exit(0);
}
```

添加编译路径到make

```
//路径 makefile

UPROGS=\
    $_cat\
    $_echo\
    $_forktest\
    $_grep\
    $_init\
    $_kill\
    $_ln\
    $_ls\
    $_mkdir\
    $_rm\
    $_sh\
    $_stressfs\
    $_usertests\
    $_grind\
    $_wc\
    $_zombie\
    $_sleep\
```

```
UPROGS=\
    $U/_cat\
    $U/_echo\
    $U/_forktest\
    $U/_grep\
    $U/_init\
    $U/_kill\
    $U/_ln\
    $U/_ls\
    $U/_mkdir\
    $U/_rm\
    $U/_sh\
    $U/_stressfs\
    $U/_usertests\
    $U/_grind\
    $U/_wc\
    $U/_zombie\
    $U/_sleep\
```

编译运行

```
make clean
```

开启qemu
`make qemu`

测试一下，不测也可以
`sleep 1`

退出qemu

```
ctrl-a x
```

测试sleep

```
./grade-lab-util sleep
```

```
== Test sleep, no arguments == sleep, no arguments: OK (4.2s)
== Test sleep, returns == sleep, returns: OK (0.8s)
== Test sleep, makes syscall == sleep, makes syscall: OK (0.9s)
```

如图，则成功